

Hitesh Labh

+977 9842028611 | hiteshlabh307@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Backend developer and Electronics, Communication and Information Engineer who builds secure REST APIs with Python, FastAPI and async SQLAlchemy. Designs layered controller–service–repository backends with JWT authentication, per-user data isolation and Alembic-managed PostgreSQL schemas, and has shipped on-device TensorFlow Lite models in real-time apps. Seeking a backend engineering internship.

PROJECTS

Task Manager REST API | FastAPI, async SQLAlchemy, PostgreSQL, Alembic, Pydantic, JWT

- Designed and built a RESTful task-management API in FastAPI using a controller–service–repository architecture, separating routing, business logic and data access into independent layers.
- Implemented JWT bearer authentication with ownership-scoped queries, enforcing per-user data isolation on every read, update and delete.
- Modeled the PostgreSQL schema with async SQLAlchemy and managed versioned schema changes through Alembic migrations.
- Added partial updates (PATCH), UUID-based resource IDs and Pydantic request/response validation, and verified every endpoint through Postman.
- Debugged errors across the controller, service and repository layers by reading Python tracebacks and tracing each issue to its root cause.

Real-Time Sign Language Translator | Python, TensorFlow, TensorFlow Lite, Kotlin, CameraX

- Built an end-to-end system that translates American Sign Language into text and speech in real time, to support communication for deaf-mute users.
- Collected a custom sign language dataset using smartphone cameras (at least 25 frames per sign) and trained two 1D CNN models on it, one for the alphabet and one for a 50-word vocabulary, with normalization preprocessing.
- Deployed both models on-device by converting them to TensorFlow Lite and integrating them into a native Kotlin Android app, so recognition runs without a server.
- Implemented a live CameraX capture pipeline that shows recognized signs as on-screen text and reads them aloud with text-to-speech.

Smart Face Attendance System | Python, Flask, SQLite, FaceNet, RFID, JavaScript

- Built a dual-mode attendance system that identifies users through either RFID card scan or FaceNet face recognition, using Kaggle face datasets.
- Developed a Flask REST backend handling authentication, attendance logging and RFID event processing.
- Designed a SQLite schema for user profiles, RFID-to-user mappings and timestamped attendance records.
- Built the attendance dashboard in HTML, CSS and JavaScript.

TECHNICAL SKILLS

Languages: Python, SQL, C++, C, JavaScript

Backend: FastAPI, async SQLAlchemy, Alembic, Pydantic, REST API design, JWT authentication

Databases: SQL (joins, subqueries, CTEs, CASE WHEN, GROUP BY / HAVING, COUNT DISTINCT, window functions: ROW_NUMBER, RANK, DENSE_RANK)

Machine Learning: TensorFlow, TensorFlow Lite, OpenCV, Scikit-Learn, NumPy, Pandas, Matplotlib

Frontend & Mobile: HTML, CSS, Kotlin, Android Studio, CameraX

Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook, Basic Debugging

EDUCATION

Tribhuvan University, IOE, Purwanchal Campus

Bachelor of Electronics, Communication and Information Engineering

2022 – 2026

Dharan, Nepal